1. Show student name with department name.

```
mysql> SELECT s.name, d.dept_name
    -> FROM Student s
    -> INNER JOIN Department d
    -> ON s.dept_id = d.dept_id;
+-------+-----------+
| name  | dept_name |
+-------+-----------+
| Amit  | CSE       |
| Riya  | CSE       |
| Sohan | ECE       |
| Priya | ECE       |
| Neha  | ME        |
+-------+-----------+
5 rows in set (0.01 sec)
```

2. List all students with courses.

```
mysql> SELECT s.name, c.course_name
    -> FROM Student s
    -> JOIN Enrollment e ON s.student_id = e.student_id
    -> JOIN Course c ON e.course_id = c.course_id;
+-------+------------------+
| name  | course_name      |
+-------+------------------+
| Amit  | DBMS             |
| Riya  | DBMS             |
| Amit  | AI               |
| Sohan | Digital Circuits |
| Priya | Digital Circuits |
| Neha  | Thermodynamics   |
+-------+------------------+
6 rows in set (0.01 sec)
```

3. Display student, course and marks.

```
mysql> SELECT s.name, c.course_name, e.marks
    -> FROM Student s
    -> JOIN Enrollment e ON s.student_id = e.student_id
    -> JOIN Course c ON e.course_id = c.course_id;
+-------+------------------+-------+
| name  | course_name      | marks |
+-------+------------------+-------+
| Amit  | DBMS             |    85 |
| Riya  | DBMS             |    90 |
| Amit  | AI               |    78 |
| Sohan | Digital Circuits |    72 |
| Priya | Digital Circuits |    88 |
| Neha  | Thermodynamics   |    65 |
+-------+------------------+-------+
6 rows in set (0.00 sec)
```

4. Show courses by department.

```
mysql> SELECT d.dept_name, c.course_name
    -> FROM Department d
    -> JOIN Course c ON d.dept_id = c.dept_id;
+-----------+------------------+
| dept_name | course_name      |
+-----------+------------------+
| CSE       | DBMS             |
| CSE       | AI               |
| ECE       | Digital Circuits |
| ME        | Thermodynamics   |
| Civil     | Surveying        |
+-----------+------------------+
5 rows in set (0.00 sec)
```

5. Show all students even without department.

```
mysql> SELECT s.name, d.dept_name
    -> FROM Student s
    -> LEFT JOIN Department d
    -> ON s.dept_id = d.dept_id;
+-------+-----------+
| name  | dept_name |
+-------+-----------+
| Amit  | CSE       |
| Riya  | CSE       |
| Sohan | ECE       |
| Neha  | ME        |
| Rahul | NULL      |
| Priya | ECE       |
+-------+-----------+
6 rows in set (0.00 sec)
```

6. Show all students even without enrollment.

```
mysql> SELECT s.name, c.course_name
    -> FROM Student s
    -> LEFT JOIN Enrollment e ON s.student_id = e.student_id
    -> LEFT JOIN Course c ON e.course_id = c.course_id;
+-------+------------------+
| name  | course_name      |
+-------+------------------+
| Amit  | DBMS             |
| Amit  | AI               |
| Riya  | DBMS             |
| Sohan | Digital Circuits |
| Neha  | Thermodynamics   |
| Rahul | NULL             |
| Priya | Digital Circuits |
+-------+------------------+
7 rows in set (0.00 sec)
```

7. Show all departments even if no students.

```
mysql> SELECT d.dept_name, s.name
    -> FROM Department d
    -> LEFT JOIN Student s
    -> ON d.dept_id = s.dept_id;
+-----------+-------+
| dept_name | name  |
+-----------+-------+
| CSE       | Amit  |
| CSE       | Riya  |
| ECE       | Sohan |
| ECE       | Priya |
| ME        | Neha  |
| Civil     | NULL  |
+-----------+-------+
6 rows in set (0.00 sec)
```

8. Show all courses even if no enrollment.

```
mysql> SELECT c.course_name, e.student_id
    -> FROM Course c
    -> LEFT JOIN Enrollment e
    -> ON c.course_id = e.course_id;
+------------------+------------+
| course_name      | student_id |
+------------------+------------+
| DBMS             |        101 |
| DBMS             |        102 |
| AI               |        101 |
| Digital Circuits |        103 |
| Digital Circuits |        106 |
| Thermodynamics   |        104 |
| Surveying        |       NULL |
+------------------+------------+
7 rows in set (0.00 sec)
```

9. Full join students and departments.

```
mysql> SELECT s.name, d.dept_name
    -> FROM Student s
    -> LEFT JOIN Department d ON s.dept_id = d.dept_id
    ->
    -> UNION
    ->
    -> SELECT s.name, d.dept_name
    -> FROM Student s
    -> RIGHT JOIN Department d ON s.dept_id = d.dept_id;
+-------+-----------+
| name  | dept_name |
+-------+-----------+
| Amit  | CSE       |
| Riya  | CSE       |
| Sohan | ECE       |
| Neha  | ME        |
| Rahul | NULL      |
| Priya | ECE       |
| NULL  | Civil     |
+-------+-----------+
7 rows in set (0.01 sec)
```

10. Show student, department, course, marks.

```
mysql> SELECT s.name, d.dept_name, c.course_name, e.marks
    -> FROM Student s
    -> JOIN Department d ON s.dept_id = d.dept_id
    -> JOIN Enrollment e ON s.student_id = e.student_id
    -> JOIN Course c ON e.course_id = c.course_id;
+-------+-----------+------------------+-------+
| name  | dept_name | course_name      | marks |
+-------+-----------+------------------+-------+
| Amit  | CSE       | DBMS             |    85 |
| Riya  | CSE       | DBMS             |    90 |
| Amit  | CSE       | AI               |    78 |
| Sohan | ECE       | Digital Circuits |    72 |
| Priya | ECE       | Digital Circuits |    88 |
| Neha  | ME        | Thermodynamics   |    65 |
+-------+-----------+------------------+-------+
6 rows in set (0.00 sec)
```

11. CSE students scoring above 80.

```
mysql> SELECT s.name, e.marks
    -> FROM Student s
    -> JOIN Department d ON s.dept_id = d.dept_id
    -> JOIN Enrollment e ON s.student_id = e.student_id
    -> WHERE d.dept_name = 'CSE' AND e.marks > 80;
+------+-------+
| name | marks |
+------+-------+
| Amit |    85 |
| Riya |    90 |
+------+-------+
2 rows in set (0.00 sec)
```

12. Count students per department.

```
mysql> SELECT d.dept_name, COUNT(s.student_id) AS total_students
    -> FROM Department d
    -> LEFT JOIN Student s ON d.dept_id = s.dept_id
    -> GROUP BY d.dept_name;
+-----------+----------------+
| dept_name | total_students |
+-----------+----------------+
| CSE       |              2 |
| ECE       |              2 |
| ME        |              1 |
| Civil     |              0 |
+-----------+----------------+
4 rows in set (0.01 sec)
```

13. Average marks per course.

```
mysql> SELECT c.course_name, AVG(e.marks) AS avg_marks
    -> FROM Course c
    -> JOIN Enrollment e ON c.course_id = e.course_id
    -> GROUP BY c.course_name;
+------------------+-----------+
| course_name      | avg_marks |
+------------------+-----------+
| DBMS             |   87.5000 |
| AI               |   78.0000 |
| Digital Circuits |   80.0000 |
| Thermodynamics   |   65.0000 |
+------------------+-----------+
4 rows in set (0.00 sec)
```

14. Count students per course.

```
mysql> SELECT c.course_name, COUNT(e.student_id) AS total_students
    -> FROM Course c
    -> LEFT JOIN Enrollment e ON c.course_id = e.course_id
    -> GROUP BY c.course_name;
+------------------+----------------+
| course_name      | total_students |
+------------------+----------------+
| DBMS             |              2 |
| AI               |              1 |
| Digital Circuits |              2 |
| Thermodynamics   |              1 |
| Surveying        |              0 |
+------------------+----------------+
5 rows in set (0.00 sec)
```

15. Courses with more than 1 student.

```
mysql> SELECT c.course_name, COUNT(e.student_id) AS total
    -> FROM Course c
    -> JOIN Enrollment e ON c.course_id = e.course_id
    -> GROUP BY c.course_name
    -> HAVING COUNT(e.student_id) > 1;
+------------------+-------+
| course_name      | total |
+------------------+-------+
| DBMS             |     2 |
| Digital Circuits |     2 |
+------------------+-------+
2 rows in set (0.01 sec)
```

16. Departments with more than 2 students.

```
mysql> SELECT d.dept_name, COUNT(s.student_id) AS total
    -> FROM Department d
    -> JOIN Student s ON d.dept_id = s.dept_id
    -> GROUP BY d.dept_name
    -> HAVING COUNT(s.student_id) > 2;
Empty set (0.00 sec)
```

17. Students from same city.

```
mysql> SELECT s1.name, s2.name, s1.city
    -> FROM Student s1
    -> JOIN Student s2
    -> ON s1.city = s2.city AND s1.student_id <> s2.student_id;
+-------+-------+-------+
| name  | name  | city  |
+-------+-------+-------+
| Sohan | Amit  | Delhi |
| Priya | Riya  | Noida |
| Amit  | Sohan | Delhi |
| Riya  | Priya | Noida |
+-------+-------+-------+
4 rows in set (0.01 sec)
```

18. All student-course pairs.

```
mysql> SELECT s.name, c.course_name
    -> FROM Student s
    -> CROSS JOIN Course c;
+-------+------------------+
| name  | course_name      |
+-------+------------------+
| Amit  | Surveying        |
| Amit  | Thermodynamics   |
| Amit  | Digital Circuits |
| Amit  | AI               |
| Amit  | DBMS             |
| Riya  | Surveying        |
| Riya  | Thermodynamics   |
| Riya  | Digital Circuits |
| Riya  | AI               |
| Riya  | DBMS             |
| Sohan | Surveying        |
| Sohan | Thermodynamics   |
| Sohan | Digital Circuits |
| Sohan | AI               |
| Sohan | DBMS             |
| Neha  | Surveying        |
| Neha  | Thermodynamics   |
| Neha  | Digital Circuits |
| Neha  | AI               |
| Neha  | DBMS             |
| Rahul | Surveying        |
| Rahul | Thermodynamics   |
| Rahul | Digital Circuits |
| Rahul | AI               |
| Rahul | DBMS             |
| Priya | Surveying        |
| Priya | Thermodynamics   |
| Priya | Digital Circuits |
| Priya | AI               |
| Priya | DBMS             |
+-------+------------------+
30 rows in set (0.00 sec)
```

19. Top scorer in each course.

```
mysql> SELECT c.course_name, s.name, e.marks
    -> FROM Enrollment e
    -> JOIN Student s ON e.student_id = s.student_id
    -> JOIN Course c ON e.course_id = c.course_id
    -> WHERE (e.course_id, e.marks) IN (
    ->     SELECT course_id, MAX(marks)
    ->     FROM Enrollment
    ->     GROUP BY course_id
    -> );
+------------------+-------+-------+
| course_name      | name  | marks |
+------------------+-------+-------+
| AI               | Amit  |    78 |
| DBMS             | Riya  |    90 |
| Thermodynamics   | Neha  |    65 |
| Digital Circuits | Priya |    88 |
+------------------+-------+-------+
4 rows in set (0.01 sec)
```

20. Department with highest average marks.

```
mysql> SELECT d.dept_name, AVG(e.marks) AS avg_marks
    -> FROM Department d
    -> JOIN Student s ON d.dept_id = s.dept_id
    -> JOIN Enrollment e ON s.student_id = e.student_id
    -> GROUP BY d.dept_name
    -> ORDER BY avg_marks DESC
    -> LIMIT 1;
+-----------+-----------+
| dept_name | avg_marks |
+-----------+-----------+
| CSE       |   84.3333 |
+-----------+-----------+
1 row in set (0.00 sec)
```